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1. $\bar{X}=52 \quad s=10 \quad n=100$

$SE(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{10}{\sqrt{100}} = (1)$

$H_0: \mu = 50 \quad H_A: \mu \neq 50$

$z = \frac{\bar{x}-\mu_0}{SE_{\bar{x}}} = \frac{52-50}{1} = (2)$

$\alpha = 0.10$; Reject H_0 since $z = 2 > 1.645$

$\alpha = 0.05$; Reject H_0 since $z = 2 > 1.96$

$\alpha = 0.01$; fail to reject H_0 since $z = 2 < 2.576$

95% CI: $[\bar{x} - 1.96 SE(\bar{x}), \bar{x} + 1.96 SE(\bar{x})]$

$[50.04, 53.96]$

2. $\hat{\beta} = 3,1$ $\widehat{SE}(\hat{\beta}) = 1,3$ $t = \frac{3}{SE(\hat{\beta})} = 2,385$

90% CI: $[\hat{\beta} - 1,645(SE(\hat{\beta})), \hat{\beta} + 1,645(SE(\hat{\beta}))]$
$[0,9615, 5,2385]$

Reject null at $\alpha = 0,10$
because $|t| > 1,645$, 0 is not in the corresponding CI.

95% CI: $[\hat{\beta} - 1,96(SE(\hat{\beta})), \hat{\beta} + 1,96(SE(\hat{\beta}))]$
$[0,552, 5,648]$

Reject null at $\alpha = 0,05$
because $|t| > 1,96$ 0 is not in the corresponding CI.

99% CI: $[\hat{\beta} - 2,576(SE(\hat{\beta})), \hat{\beta} + 2,576(SE(\hat{\beta}))]$
$[-0,249, 6,449]$

Fail to reject null hypothesis at $\alpha = 0,01$
because $|t| < 2,576$, 0 is in the corresponding CI.

Whenever 0 is in the CI we fail to reject the null.

3. $\bar{x} = 11.5 \quad s = 3 \quad n = 36$

$SE(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{3}{\sqrt{36}} = \frac{1}{2}$ $H_0: \mu = 12$ $H_A: \mu \neq 12$

12 is a plausible value for the population mean because it is one standard error away from the sample mean.

95% CI: $[\bar{x} - 2.576(SE(\bar{x})), \bar{x} + 2.576(SE(\bar{x}))]$

$[10.212, 12.788]$

We fail to reject the null because the value 12 is in the confidence interval.

4. $\bar{X}_A = 25 \quad s_A = 8 \quad n_A = 100$

$\bar{X}_B = 22 \quad s_B = 7 \quad n_B = 100$

$\bar{X}_A - \bar{X}_B = 25 - 22 = 3$

$SE = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} = \sqrt{\frac{64}{100} + \frac{49}{100}} = 1{,}063$

$Z = \frac{\bar{X}_A - \bar{X}_B}{SE} = \frac{25-22}{1{,}063} = 2{,}822$

95% CI: $[3 - 1{,}96(1{,}063),\ 3 + 1{,}96(1{,}063)]$

$[0{,}917,\ 5{,}084]$

5. $\bar{X}_A = 14 \quad s_A = 6 \quad n_A = 81$

$\bar{X}_B = 12 \quad s_B = 5 \quad n_B = 64 \quad H_0: \mu_A - \mu_B = 0$

$$SE = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} = \sqrt{\frac{36}{81} + \frac{25}{64}} = 0,914$$

$$z = \frac{14-12}{0,914} = 2,189$$

6. $\bar{X}_A = 3,2 \quad s_A = 1,5 \quad n_A = 49$

$\bar{X}_B = 2,8 \quad s_B = 1,4 \quad n_B = 64 \quad H_0: \mu_A - \mu_B = 0 \quad \alpha = 0,05$

$$SE = \sqrt{\frac{2,25}{49} + \frac{1,96}{64}} = 0,277 \qquad z = \frac{0,4}{0,277} = 1,446$$

95% CI: $[0,4 - 1,96(0,277),\ 0,4 + 1,96(0,277)]$

$[-0,142,\ 0,942]$ We are 95% confident that the true difference in averge defect scores is within the CI. Using $\alpha = 0,05$, our confidence interval includes 0. Thus, the difference in defect scores across the two factories is not statistically large relative to the noise.

7. a. The estimate proportion is 0.9331.

b. I used a seed of 42 and 2000 bootstrap replications.

c. The bootstrap standard error is 0.0046

d.

90% CI: [0.9256, 0.9407]
95% CI: [0.9241, 0.9421]
99% CI: [0.9213, 0.9449]

e. Using the 90% CI and $\alpha = 0.10$, we reject the null hypothesis $H_0: p = 0.90$ because 0.90 is not in the CI. We have significant statistical evidence that the true probability that a home has central air is not equal to 0.9. Using 95% and 99% CIs with α values of 0.05 and 0.01, respectively, we come to the same conclusion because 0.9 is in neither confidence interval.

8. a. I used a seed of 42 and 2000 bootstrap replications.

b. The bootstrap standard error is 0.0103

c. The estimated correlation is 0.7232.

90% CI: [0.706, 0.740]
95% CI: [0.703, 0.743]
99% CI: [0.697, 0.750]

d. We reject the null: $H_0: \rho = 0$ at all three significance levels $\alpha = 0.10, 0.05, 0.01$. This is because none of the matching CIs include zero. This means at all three significance levels, we have significant statistical evidence that there is a nonzero correlation between arcsinh of area and arcsinh of price.

e. They need to remain paired because otherwise we wouldn't be resampling real data points.

9. a. I used a seed of 42 and 2000 bootstrap replications.

b. The bootstrap standard error is 2,1358 months.

c. The original estimated mean is 129,094 months, or 10,758 years

90% CI: [125,581, 132,608] months
95% CI: [124,908, 133,280] months
99% CI: [123,592, 134,596] months

90% CI: [10,465, 11,051] years
95% CI: [10,409, 11,107] years
99% CI: [10,299, 11,216] years

d. No cafidence intervals include 120 months. Therefore, we reject the null: $H_0: \mu = 120$ months for the corresponding α values of 0.10, 0.05, and 0.01.

e. Our estimate of 129,094 months or 10,758 years is the mean recorded time to death or follow-up in the dataset

Our 90% CI tells us that we are 90% confident that the true mean time to follow-up or death is between 125,581 and 132,608 months, and between 10,465 and 11,051 years.

The other confidence intervals tell us the range of values that we are 95% and 99% confident that the true mean falls in.

These results do not estimate unrestricted mean lifetime because many people were likely still living at the time the study ended.

10. a. I used a seed of 42 and 2000 bootstrap replications.

b. The bootstrap standard error is 0,023 hou

c. The original estimated mean is 7,621 hou

90% CI: [7,584, 7,659] hours
95% CI: [7,577, 7,666] hours
99% CI: [7,563, 7,680] hours

d. H_0: $\mu = 8$ hours

None of the confidence intervals include 8 hours, so we reject the null at the significance levels $\alpha = 0,10$, 0,05, 0,01.

e. Our results show that we have significant statistical evidence, at all three of those significance levels, that the true mean sleep is not equal to 8 hours. However, in practical terms, our estimated mean of 7,621 hours is only 21 minutes less, which might not be considered a large difference.